

Dylan Ong

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EDUCATION

University of Wisconsin-Madison | Madison, WI

May 2026

Bachelor of Science, Double Major: Computer Science & Data Science

GPA: 3.93/4.0

- Certificate (Minor): Accounting Fundamentals
- Honors: Dean's List for all semesters
- Relevant Coursework: Data Structures & Algorithms, Data Science Modeling, Operating Systems, Artificial Intelligence, Machine Learning, Optimization, Big Data, Database Management, Finance, Risk Management

SKILLS

- **Programming:** Python, R, SQL, Java, C++, C
- **App Development:** React, Node.js, JavaScript, HTML/CSS
- **Data Science & ML:** Pandas, NumPy, scikit-learn, PyTorch, TensorFlow, Neural Networks
- **Frameworks & Tools:** Git, Docker, Bash, Unix, CI/CD, REST APIs, Visual Studio, Streamlit, Shiny
- **Data Platforms & Distributed Systems:** Kafka, Cassandra, Spark, Hadoop, BigQuery
- **Languages:** Bahasa Indonesia (native), English (fluent), Mandarin Chinese (conversational)

WORK EXPERIENCE

Senior Research Intern for Development | Center for Research on Complex Thinking September 2025 - May 2026

- **Managed** a team of developers in building and maintaining educational research platforms: Epistemic Network Analysis (data analysis/visualization), iPlan (land-use simulation), and Codey (LLM-based qualitative coding).
- **Led quality assurance testing** for educational research projects, coordinating researchers and developers to **identify bugs** and maintain consistent platform performance.
- Designed and implemented a workflow process that **improved team efficiency**.
- **Provided technical support** to institutional users, **troubleshooting** platform issues and **debugging** errors

Data Analyst Research Intern | Center for Research on Complex Thinking

October 2024 - September 2025

- Performed **quality assurance testing** on Epistemic Network Analysis, iPlan, and Codey platforms.
- Conducted **data analysis** on various education-related projects using **R** and **Epistemic Network Analysis**.
- Provided technical and analytical support for **30+ researchers** at the QE Fellows Institute.

Student Stocker | University of Wisconsin-Madison Housing | Madison, WI

February 2023 - September 2023

- Coordinated with other employees in **stocking** and **organizing** goods.
- Provided service to customers by **serving food** and **ensuring proper hygiene** in food stations.
- **Operated the POS system** to manage financial transactions with customers

PROJECTS

Rosenberg Autism App | *Fullstack Development* | Python, SQLite, Multithreading, REST APIs, Data Processing, Kivy

- Designed and built **multithreaded architecture** for real-time eye-tracking experiments using **Python**
- Integrated application with Pupil Labs **eye tracker API** to stream and process gaze data in real time
- Developed interactive GUI using **Kivy** to manage experiment flow
- Implemented **data pipeline** and **SQLite** database for storing session data and experiment results

Financial Stock Analysis | *Fullstack Development* | Python, TensorFlow/Keras, Scikit-learn, Flask, React, REST APIs

View project on: <https://financial-stock-analysis.onrender.com/>

- Built a multi-stock **LSTM model** in **Python** to predict stock prices using historical data and scaling techniques
- Developed a **Flask API** backend to serve real-time predictions and historical stock data
- Created a responsive **React** frontend with interactive charts to visualize historical prices and predicted trends

Strava Analyzer | *Fullstack Development* | *Python, Scikit-learn, Streamlit, REST APIs, OAuth*

View project on: <https://strava-analyzers.streamlit.app/>

- Developed a **machine learning pipeline** in **Python** to predict running pace using **Random Forest regression**
- Built a **Streamlit web app** enabling dynamic input of planned runs and **visualization** of past performance trends
- Implemented **Strava API integration**, including **OAuth** token refresh, to automate activity retrieval

MiniSpark Project | *Operating Systems Project* | *C, Data Processing, Concurrency*

- Built a simplified **Spark-like data processing engine** using **RDD** abstractions in **C**
- Implemented **multithreaded** task execution with synchronization via **locks** and **condition variables**
- Designed **concurrent workflows** for parallel data processing

COVID-19 Mortality Rates Analysis | *Data Analysis* | *R, RStudio, Data Visualization, Statistical Analysis*

- **Collected data** pertaining to COVID-19 cases for the general population and prison facilities in the Midwest
- Analyzed and visualized mortality rate differences in **RStudio**, verified results with **hypothesis testing**
- Found that mortality rates were ~0.014 higher in the general population, confirming a significant difference

Madison Campus Route Seeking Program | *Fullstack Development* | *Java, JavaFX, GUI Development*

- Implemented path-seeking program using Dijkstra's Algorithm in **Java**
- Designed app frontend using **JavaFX**
- Integrated frontend and backend codes into a complete application

LEADERSHIP AND INVOLVEMENT

General Member | Circle K International | Madison, WI

September 2023 - September 2024

- Volunteered at a food pantry by assisting its team members in preparing and serving food to those in need
- Collaborated together with fellow members to decorate a homeless shelter
- Participated in various community service projects

Volunteer | Madison Children's Museum | Madison, WI

September 2023 - September 2024

- Interacted and played with children in the museum
- Supported museum staff in preparing various playing areas and events
- Guide and assist museum visitors in navigating the building

CERTIFICATIONS

Completion of Apple Foundation Program | Apple Developer Academy | Surabaya, Indonesia

June - July 2019

Certificate of Chinese Proficiency (HSK 4) | Xin Zhong Course School | Surabaya, Indonesia

July 2010 - May 2021